Nithin Sai Ega

eganithinsai@gmail.com | linkedin.com/in/nithin-sai-ega | github.com/nithin-sai-ega

Education

Rutgers University, School of Arts and Sciences New Brunswick, NJ
B.S. in Computer Science; Second Major in Data Science; Minor in Mathematics *Expected May 2027*
GPA: 4.0 / 4.0

Relevant Coursework: Design & Analysis of Computer Algorithms, Artificial Intelligence, Deep Learning, Data Structures, Differential Equations

Technical Projects

EchoTouch: Touchless Gesture Control via Acoustic Doppler github.com/nithin-sai-ega/EchoTouch
Python, PyTorch, NumPy/SciPy, Digital Signal Processing, Streamlit

- Built a camera-free gesture recognition system that turns a laptop's built-in speaker and microphone into a sonar sensor, detecting hand motion from Doppler shifts in a near-ultrasonic (17–21 kHz) tone.
- Designed a real-time DSP pipeline (STFT, carrier-band cropping, static background subtraction, normalization) feeding a custom PyTorch CNN, reaching **96% accuracy** on a held-out recording session across 4 gesture classes.
- Prevented data leakage with session-level train/validation splits; added class-balanced sampling, Doppler-safe augmentation, and early stopping.
- Engineered a low-latency multithreaded audio engine with a thread-safe ring buffer, plus a motion gate, temporal smoothing, and cooldown logic to reduce false triggers.
- Developed a physics-based acoustic simulator and **124 automated tests** to validate the signal-processing and gesture-detection logic.
- Shipped a Streamlit dashboard with live Doppler spectrograms, guided data collection, background training, and hardware diagnostics that auto-calibrate each device.

Course Snipe Web App *Spring Boot, React, Vite, Material UI*

- Developing a full-stack web application that monitors and displays open course sections in real time for Rutgers students.
- Designed RESTful backend endpoints to fetch, filter, and process course data efficiently.
- Built an interactive, responsive frontend with React and Material UI to streamline student class registration.

Skills

Languages: Python, Java, C++, SQL, HTML/CSS, LaTeX, R

ML & Data: PyTorch, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Plotly, CNNs, Digital Signal Processing

Software & Tools: Spring Boot, React, Node.js, Streamlit, REST APIs, Git, pytest, VS Code, Postman, PostgreSQL, MySQL

Campus Involvement

Undergraduate Student Alliance of Computer Science (USACS) *Active Member*
Rutgers Data Science Club *Active Member*